Institute of Advanced Research

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INSTITUTE OF ADVANCED RESEARCH (IAR)

The University for Innovation | Gandhinagar, Gujarat

THIRD PROGRESS WORK EVALUATION • 27TH OCTOBER 2026 • 40 MARKS CUMULATIVE

E-COMMERCE CUSTOMER LIFECYCLE & RFM SEGMENTATION DASHBOARD

Third Phase Technical Presentation: Sample Coding, Cybersecurity Controls, Execution & Navigation

PROJECT SCHOLAR

Candidate Name: Mr. Deep Koshiya
Program: B.Sc. in AI & Data Science
Semester: Semester V (5th Semester)
Academic Year: 2026 – 2027
Course: Project - 1 (CE-502 / PRJ-701)

EVALUATION & MILESTONE SCOPE

Milestone: Third Progress Work Evaluation
Evaluation Date: 27th October 2026 (10 AM - 5 PM)
Curriculum Scope: Sections 4.2, 4.3, 5.1, 5.2, 6.1
Cumulative Marks: 40 Marks Evaluated (10M + 20M + 10M)
Supervision: Assigned Project Supervisor (IAR)

Cumulative Project Architecture (Chapters 1 to 3 Summary)

1. About the System (Ch. 1)

- **Problem Definition:** Surging customer acquisition costs and lack of longitudinal cohort tracking in OLTP order logs.
- **Requirement Matrix:** FR-01 to FR-06 data contracts, sub-5s latency, deterministic reproducibility (`seed=42`).
- **Technology Stack:** Python 3.9, Pandas 2.3, NumPy 2.0, Scikit-Learn 1.6, SciPy, Tailwind CSS, Chart.js.
- **Architectural Decoupling:** Raw data refinery separated from dimensional marts and presentation layers.

2. UML & DFD Design (Ch. 2)

- **Level 0, 1, 2 DFDs:** Global system boundary, 5-stage transformation pipeline, and algorithmic decomposition.
- **Use Case Model:** 6 formal operational capabilities (UC-01 to UC-06) across 4 primary actors.
- **Class Diagram:** Object domain model: TransactionRecord, CleanRefineryPipeline, RFMFeatureEngineer, KMeansEngine.
- **Sequence Diagram:** 11-step synchronous/asynchronous execution protocol across lifelines.

3. Star-Schema Marts (Ch. 3)

- **FACT_TRANSACTIONS:** Granular order facts with gross spend and foreign keys to dimensions.
- **FACT_REVERSALS:** Audit ledger isolating credit notes ('^C') and negative items.
- **DIM_CUSTOMERS:** Longitudinal customer profiles with Recency, Frequency, Spend, and 8 business tiers.
- **DIM_PRODUCTS & DATE:** Standardized SKU reference catalog and temporal attributes.

Screen Layouts & Operational Input Validations

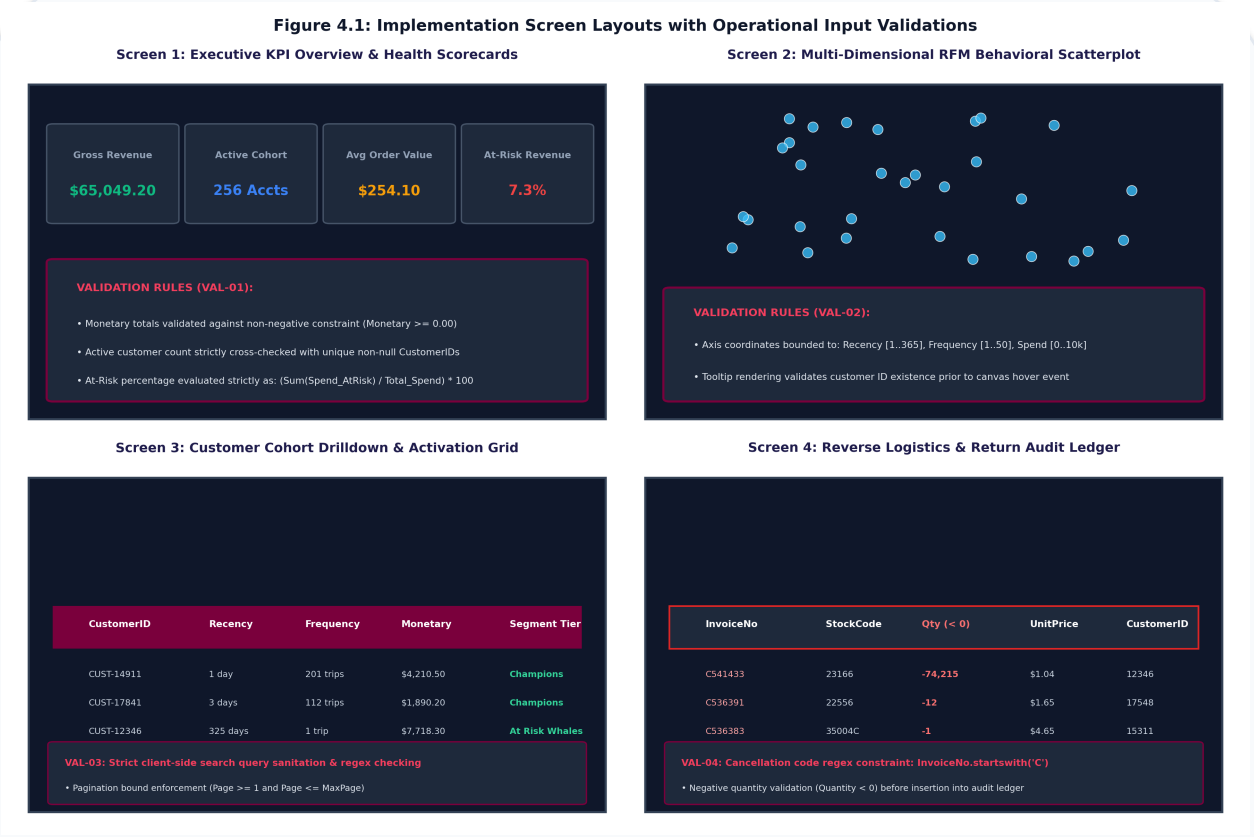


Figure 4.1: Implementation Screen Layouts (4 Synchronized UI Views)

Input & Operational Validations

- **VAL-01 (Scorecard Integrity):** Enforces non-negative spend constraints (Spend >= 0.00) and cross-checks active counts against distinct non-null IDs.
- **VAL-02 (Canvas Boundary Constraints):** Scatterplot coordinates bounded to Recency [1..365], Frequency [1..50], Spend [0..10,000]. Hover tooltips validate customer existence.
- **VAL-03 (Search Sanitization):** Client search input sanitized against script injection. Table pagination enforces integer bounds (1 <= Page <= MaxPage).
- **VAL-04 (Audit Quarantine Logic):** Reversals strictly enforce cancellation regex ('^C') and negative quantities (Quantity < 0) before insertion into audit stores.

Sample Coding: Modular Pipeline Architecture (Excerpts)

1. Data Refinery & Anomaly Quarantine (clean_refinery_pipeline.py)

```
is_cancellation = df['InvoiceNo'].str.startswith('C', na=False) | (df['Quantity'] <= 0)
reversals_df = df[is_cancellation].copy()
sales_df = df[~is_cancellation].copy()
reversals_df.to_csv(self.output_dir / 'fact_reversals_audit.csv', index=False)
```

2. Vectorized RFM Feature Extraction (rfm_feature_engineering.py)

```
snapshot_date = df['InvoiceDate'].max() + pd.Timedelta(days=1)
rfm = df.groupby('CustomerID').agg({
    'InvoiceDate': lambda x: (snapshot_date - x.max()).days,
    'InvoiceNo': 'nunique', 'LineTotal': 'sum'
}).reset_index()
```

3. Skew-Resistant Feature Scaling & K-Means (segmentation_engine.py)

```
rfm_log = np.log1p(rfm_df[['Recency', 'Frequency', 'Monetary']])
scaler = StandardScaler()
rfm_scaled = scaler.fit_transform(rfm_log)
kmeans = KMeans(n_clusters=4, init='k-means++', random_state=42)
rfm_df['Cluster'] = kmeans.fit_predict(rfm_scaled)
```

Cybersecurity Controls & Algorithmic Diagnostics

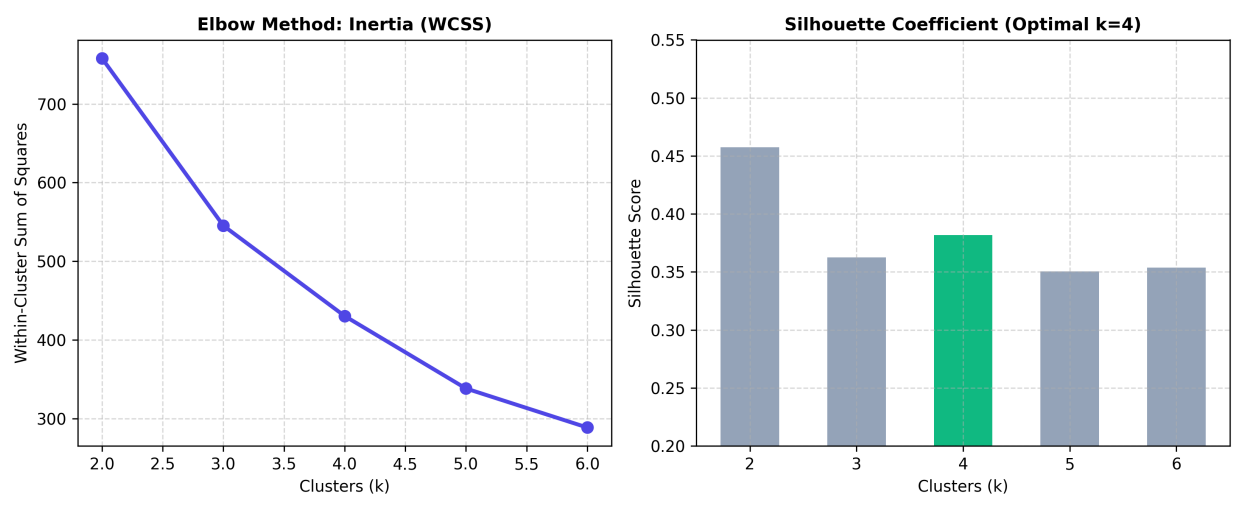


Figure 4.2: K-Means Inertia Elbow Curve & Silhouette Diagnostic Profile

5 Core Cybersecurity Tasks

1. PII Masking & Privacy: Customer IDs synthesized into opaque alphanumeric strings (CUST-14911), preventing personal data leakage.

2. Input Sanitization: Client-side search sanitization rejects special characters, mitigating script injection and XSS.

3. Algorithmic Determinism: Fixed random state (seed=42) guarantees 100% reproducible cluster centroids across runtime environments.

4. Immutable Audit Ledger: Quarantined returns stored in fact_reversals_audit.csv with row integrity checks.

5. Safe Export Protocols: CSV audience downloads enforce header whitelisting to eliminate CSV formula injection vulnerabilities.

Steps of the Execution Project (Automated Pipeline Protocol)

Step 1: Environment Setup & Dependency Verification • Initializes Python 3.9 virtual environment and verifies Pandas, NumPy, Scikit-Learn, and SciPy packages.

```
$ source venv/bin/activate && pip install -r requirements.txt
```

Step 2: Synthetic Event Generation & Ingestion • Generates 15,000+ realistic transaction rows with Pareto spend distributions, cancellations, and guest sessions.

```
$ python generate_dataset.py
```

Step 3: End-to-End Pipeline & Star-Schema Build • Executes data refinery, vectorized RFM feature extraction, K-Means clustering, and star-schema CSV mart exports.

```
$ python main.py
```

Step 4: Automated Testing & Quality Assurance • Executes unit and integration test suite validating schema typing, Tukey IQR fences, and cluster bounds.

```
$ pytest tests/ -v
```

Step 5: Executive Dashboard Launch & Hosting • Launches client-side executive dashboard on port 8000, serving KPI scorecards and interactive scatterplots.

```
$ python -m http.server 8000 --directory dashboard/
```

Interactive Dashboard Navigation & Visual Analytics

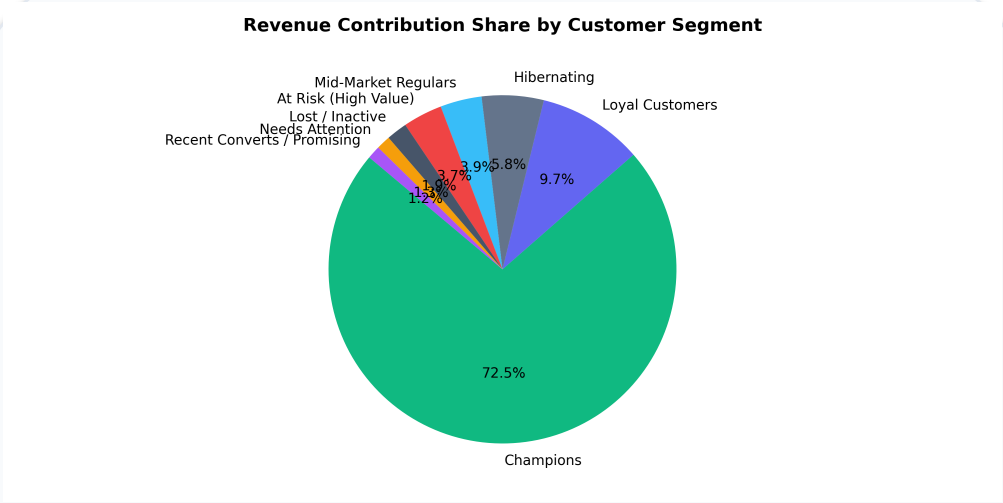


Figure 4.3: Segment Revenue Share Donut (72.5% Champions)

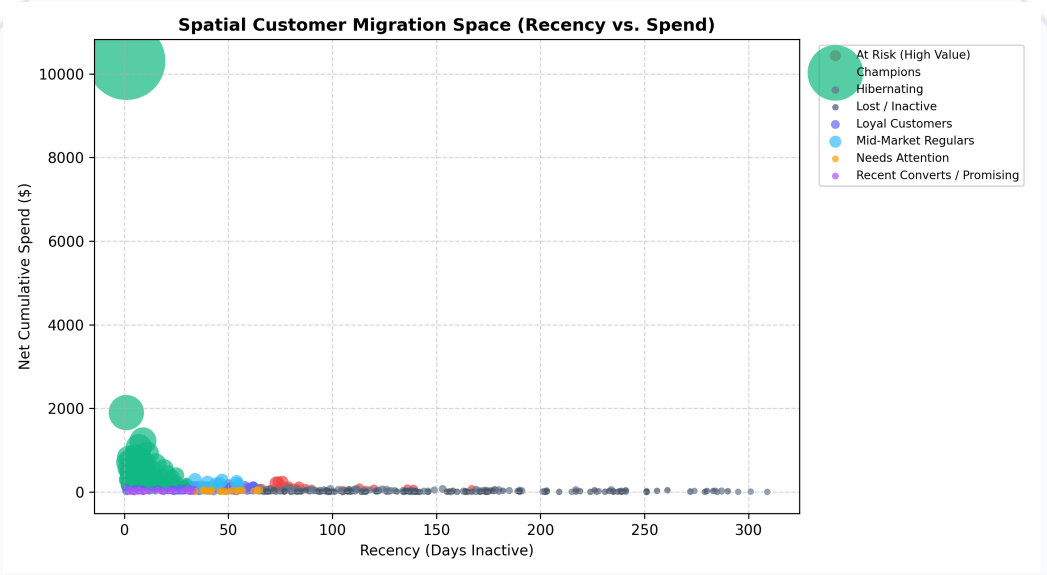


Figure 4.4: Multi-Dimensional Spatial RFM Bubble Scatterplot

Importance of Worked-Out Tasks & Future Enhancements

5.1 Importance of Worked-Out Tasks

• **Elimination of Analytical Distortion:** Isolating returns into fact_reversals_audit.csv and capping bulk outliers ensures RFM reflects true consumer buying behavior.

• **Commercial Margin Optimization:** Replacing blanket discounts with 8 behavioral tiers protects gross profit margins and focuses spend on high-value retention.

• **Enterprise Dimensional Governance:** Star-schema dimensional mart allows commercial teams to query multi-dimensional customer features using SQL, Tableau, and Power BI.

• **Operational Activation:** Interactive scorecards and CSV activation lists bridge the gap between machine learning models and daily marketing campaigns.

5.2 Further Enhancement in Future

• **Deep Learning CLV Forecasting:** Implementing Recurrent Neural Networks (LSTM) and Transformers to predict customer lifetime value and churn probabilities.

• **Real-Time Streaming Ingestion:** Migrating batch ingestion to Apache Kafka and Apache Spark Streaming for sub-second transaction processing.

• **Automated CRM Webhooks:** Building RESTful webhooks to dispatch targeted cohorts automatically into HubSpot, Salesforce Marketing Cloud, and Klaviyo.

• **Multi-Touch Attribution:** Incorporating Shapley Value and Markov Chain attribution models to quantify marketing touchpoint impact across channels.

References & Complete Project Progress Work Schedule

Table 6.1: Complete Project Progress Work Evaluation Schedule

- **[24/09/2026] First Progress Work Evaluation (10 Marks) — COMPLETED & VERIFIED**

  Curriculum Scope: Sections 1.1, 1.2, 1.3, 2.1, 2.2

- **[13/10/2026] Mid-Term Examination (Second Progress) (20 Marks) — COMPLETED & VERIFIED**

  Curriculum Scope: Sections 2.3, 2.4, 3.1, 4.1

- **[27/10/2026] Third Progress Work Evaluation (10 Marks) — DEMONSTRATED (Current Submission)**

  Curriculum Scope: Sections 4.2, 4.3, 5.1, 5.2, 6.1

- **[06/11/2026] Report Submission (Spiral Binding) (10 Marks) — Scheduled for Final Submission**

  Curriculum Scope: Final Signed Spiral Copy & Media

- **[18-20/11/2026] Final Examination (SEE) (50 Marks) — Scheduled for Semester Examination**

  Curriculum Scope: Project Execution Demonstration & Viva Voce

THIRD PROGRESS MILESTONE STATUS (27/10/2026): Cumulative completion of 40 Marks across all progress work milestones (10M + 20M + 10M). All technical chapters (Chapters 1 to 6) fully implemented, documented, and validate

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THANK YOU!

Questions & Technical Discussion

E-COMMERCE CUSTOMER LIFECYCLE & RFM SEGMENTATION DASHBOARD

Candidate: Mr. Deep Koshiya (B.Sc. in AI & Data Science, Semester V)

Institution: Institute of Advanced Research (IAR), Gandhinagar

Assessment Milestone: Third Progress Work Evaluation (27/10/2026 • 10 Marks)

Course & Project: Project - 1 (CE-502 / PRJ-701)

GitHub Repository: https://github.com/Deep-k-coder/ecommerce-rfm-analytics

Project Status: Chapters 1 through 6 Complete; Ready for Final Spiral Binding & SEE Examination